# SIMATS ENGINEERING

## ASSESSMENT TOOL 2: Scenario-Based

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| **COURSE NAME:** | Cloud Computing and Big Data Analytics | **COURSE CODE:** | CSA1522 |

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| Name | **Kanmani L** |
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**COURSE OUTCOME COVERED**

**CO2:** *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Dave's Taxonomy: Manipulation (Level 2) **Total Marks: 25**
Question Count: 5

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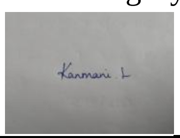
**Code of Conduct**

I ______________Kanmani L(192511414)______ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: __________ 

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**Assessment Tasks**
**Scenario-Based Questions**

1. A media company wants to migrate its video streaming platform to the cloud. The system must handle millions of concurrent users with minimal buffering. Design a cloud architecture using scalability and caching techniques. Explain how load balancing and auto-scaling can ensure smooth streaming. Discuss cost optimization strategies for such high-demand workloads.

2. A fintech startup is deploying applications using containers across multiple cloud providers. They face challenges in maintaining consistency and managing deployments. Propose a solution using container orchestration and multi-cloud strategies. Explain how service discovery and scaling can be handled efficiently. Highlight the benefits and risks of a multi-cloud deployment model.

3. An e-learning platform experiences data loss due to accidental deletion in the cloud. The organization needs a robust backup and disaster recovery strategy. Design a solution using cloud-native backup, replication, and versioning. Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved. Provide steps to ensure business continuity during failures.

4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data. Some applications must remain on-premises due to compliance requirements. Design a hybrid architecture that ensures secure communication between environments. Explain how identity management and encryption should be implemented. Discuss challenges in managing hybrid cloud environments.

5. A SaaS provider needs to ensure high availability and fault tolerance for its application. The system must continue operating even if one region goes down. Propose a multi-region deployment strategy in the cloud. Explain how data replication and failover mechanisms should be configured. Discuss monitoring and alerting strategies for proactive issue detection.

**Scenario-Based Rubric**

| Criteria | Weightage | Level 4 - Exemplary | Level 3 - Proficient | Level 2 - Developing | Level 1 - Beginning |
|---|---|---|---|---|---|
| Understanding of Scenario | 20% | Demonstrates complete understanding of the scenario context | Good understanding with minor gaps | Partial understanding | Fails to understand the scenario |
| Identification of Risks / Issues | 20% | Identifies all major issues correctly | Identifies most issues | Limited issue identification | Misses major issues |
| Application of Concepts | 25% | Applies virtualization concepts effectively | Applies concepts with minor mistakes | Partially correct application | Weak application |
| Decision Justification | 20% | Decisions are well justified and technically strong | Reasonable justification | Weak justification | No useful justification |
| Clarity | 15% | Clear and well-organized response | Generally clear | Somewhat unclear | Poorly presented |

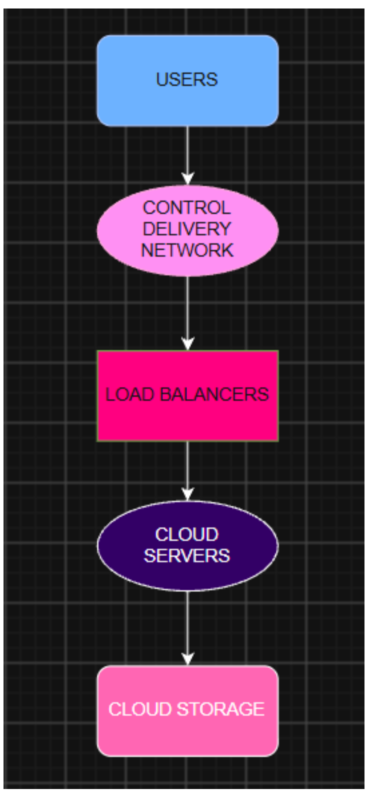
1. A media company wants to migrate its video streaming platform to the cloud. The system must handle millions of concurrent users with minimal buffering. Design a cloud architecture using scalability and caching techniques. Explain how load balancing and auto-scaling can ensure smooth streaming. Discuss cost optimization strategies for such high-demand workloads.

Answer:

A media company that provides video streaming services may have millions of users watching videos at the same time. So, the company needs a cloud system that can handle heavy traffic and provide videos without much buffering.

<u>Cloud Architecture</u>

The basic architecture can be:




- Cloud Storage is used to store all the video files.
- CDN (Content Delivery Network) keeps copies of popular videos in different locations closer to the users.
- Load Balancer distributes user requests among different servers.

- Cloud Servers handle user requests and other application activities.
- Auto-scaling adds or removes servers depending on the number of users.

Scalability

- The cloud platform should be scalable. When the number of users increases, more servers can be added automatically. When the number of users decreases, extra servers can be removed.

Caching and CDN

- Caching helps to store frequently watched videos temporarily. A CDN stores popular videos in servers that are closer to the users.

Load Balancing

- A load balancer distributes users among different servers. Instead of sending all users to one server, it divides the workload between multiple servers.

Auto-Scaling

- Auto-scaling automatically increases or decreases the number of servers based on demand.

Cost Optimization

The company can reduce cloud costs by:

- Using CDN to reduce the load on main servers.
- Using auto-scaling to avoid paying for unused servers.
- Compressing videos to reduce storage and bandwidth costs.

Thus, a cloud-based video streaming platform can use CDN, caching, load balancing, and auto-scaling to support millions of users. These techniques reduce buffering, improve performance, and keep the service available even during high traffic.

2. A fintech startup is deploying applications using containers across multiple cloud providers. They face challenges in maintaining consistency and managing deployments. Propose a solution using container orchestration and multi-cloud strategies. Explain how service discovery and scaling can be handled efficiently. Highlight the benefits and risks of a multi-cloud deployment model.

Answer:

A fintech startup may use containers to deploy its applications on different cloud platforms such as AWS, Microsoft Azure, and Google Cloud. However, managing containers across multiple clouds can be difficult because each cloud may have different services and configurations. To solve this problem, the startup can use Managing containers and a multi-cloud strategy.

<u>To Manage container</u>:

The startup can use Docker which  creates the containers, and Kubernetes manages them.The same Docker container can be deployed on different cloud platforms, which helps maintain consistency.

Kubernetes can automatically:

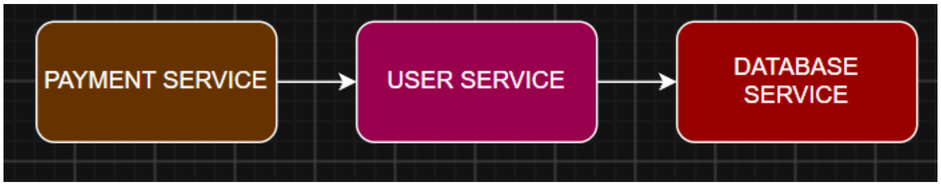
- Deploy and manage containers.
- Restart failed containers.
- Monitor the health of applications.
- Update applications with minimum downtime.
- Increase or decrease containers based on demand.

<u>Multi-Cloud Strategy</u>

The startup can run its applications on more than one cloud provider. For example, the main application can run on AWS, while another copy can run on Azure or Google Cloud.

<u>Service Discovery</u>

In a container-based application, different services need to communicate with each other. Kubernetes provides built-in service discovery using Services.




Each service has a stable name, so other services can easily find it even if containers are restarted or their IP addresses change. For communication between different clouds, secure networks, VPNs, or service mesh technologies can be used.

Scaling

- Kubernetes can automatically increase or decrease the number of containers based on the workload.
- when many users are using a payment application, more containers can be created to handle the extra traffic. When the traffic decreases, unnecessary containers can be removed.
- This helps the application remain fast and available while also avoiding unnecessary resource usage.

Benefits of Multi-Cloud

The main benefits are:

- Avoids vendor lock-in by not depending on one cloud provider.
- Improves availability because another cloud can be used if one provider fails.
- Provides flexibility to choose the best services from different providers.

Risks of Multi-Cloud

Multi-cloud also has some challenges:

- Managing multiple clouds is more complex.
- It may require more skilled professionals.
- Security management becomes more difficult.
- Connecting different cloud environments can be challenging.

Thus, the fintech startup can use Docker and Kubernetes to manage containers consistently across multiple cloud providers. Kubernetes can handle service discovery and automatic scaling, while tools like Terraform and CI/CD pipelines help maintain consistent deployments.

3. An e-learning platform experiences data loss due to accidental deletion in the cloud. The organization needs a robust backup and disaster recovery strategy. Design a solution using cloud-native backup, replication, and versioning. Explain how recovery time objective (RTO) and recovery point objective (RPO) are achieved. Provide steps to ensure business continuity during failures.

Answer:

An e-learning platform stores a lot of important information such as student details, course videos, study materials, assignments, exam marks, and payment information. If any of this data is accidentally deleted or lost, it can cause serious problems.

Cloud Backup

The organization should take regular backups of all important data and store them in cloud storage.

- Take daily or weekly backups depending on the importance of the data.
- Automatically back up databases and important files.
- Store backups separately from the main data.

If data is accidentally deleted, the organization can use the backup to recover it.

Versioning

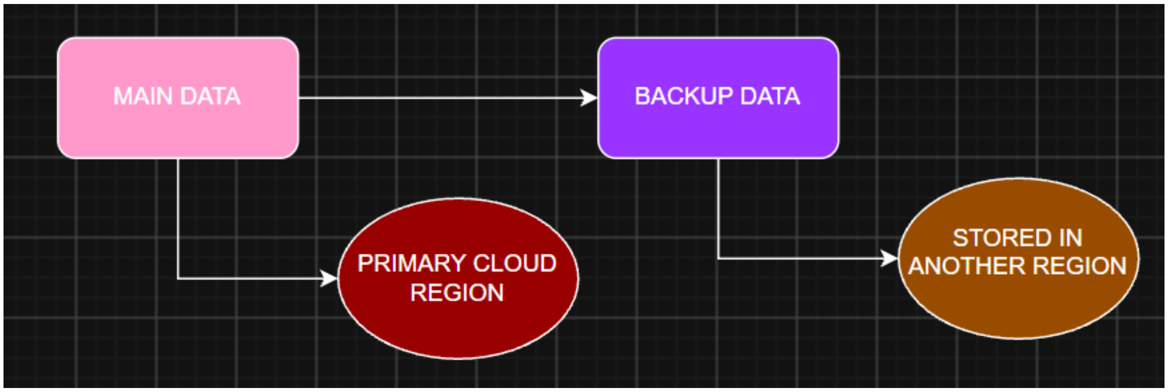
Versioning keeps different older versions of a file or data.

Versioning helps to:

- Recover accidentally deleted files.
- Restore older versions of files.
- Protect against unwanted changes.
- Keep a history of changes made to important files.

Data Replication

Replication means keeping a copy of the data in another location or cloud region.




If the main cloud region stops working because of a technical problem, natural disaster, or network failure, the organization can use the backup copy from another region.

Recovery Time Objective (RTO)

RTO means the maximum time allowed to bring the system back after a failure.

A low RTO can be achieved by:

- Keeping a backup system ready.
- Using automatic recovery.
- Keeping a copy of data in another region.
- Using automatic failover.
- Having a clear recovery plan.

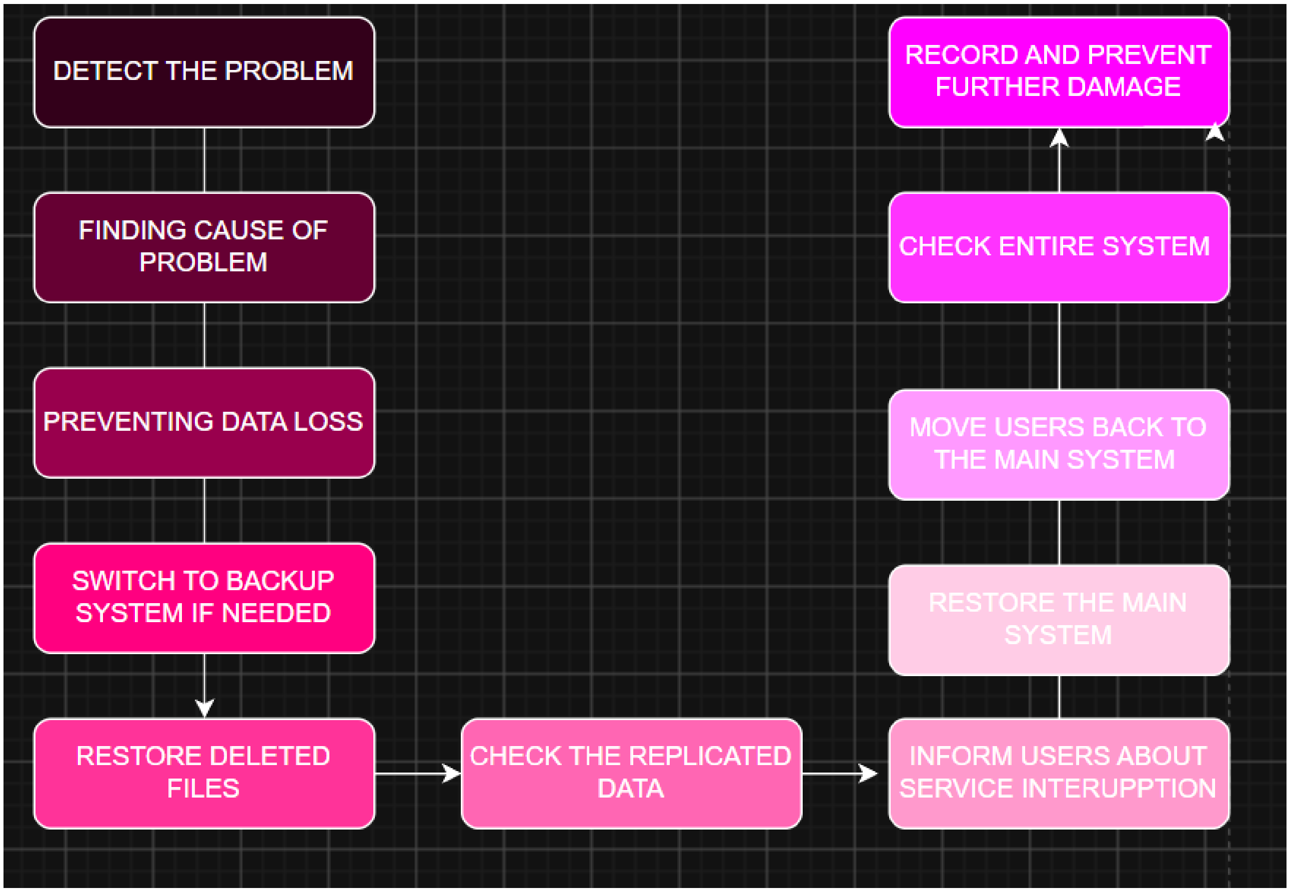
Recovery Point Objective (RPO)

RPO means the maximum amount of data that the organization can afford to lose.

A low RPO can be achieved by:

- Taking backups frequently.
- Using automatic backups.
- Using regular data replication.

Business Continuity During Failures




Regular Testing

The disaster recovery plan should not only be created but also tested regularly.

The organization should:

- Test backup restoration regularly.
- Check whether all important data is being backed up.
- Test the backup system for failures.
- Check how quickly the system can be restored.

The e-learning platform can protect its important data by using regular cloud backups, versioning, and data replication. Backups help recover lost data, versioning helps restore older files, and replication provides another copy of data in a different location.

4. A government agency is adopting a hybrid cloud model for sensitive and non-sensitive data. Some applications must remain on-premises due to compliance requirements. Design a hybrid architecture that ensures secure communication between environments. Explain how identity management and encryption should be implemented. Discuss challenges in managing hybrid cloud environments.

Answer:

A government agency handles different types of data. Some data is highly sensitive, such as citizen information and government records, while other data is less sensitive, such as public information and general applications. Since some applications must stay on the organization's own servers due to government rules and compliance requirements, a hybrid cloud model is a suitable solution.

<u>Proposed Hybrid Cloud Architecture</u>

The basic architecture can be:

- On-Premises Data Center: Stores highly sensitive data and applications that must follow strict government rules.
- Public Cloud: Used for non-sensitive data, public websites, and applications that need flexible resources.
- Secure Connection: A VPN or dedicated private connection can securely connect the on-premises data center with the cloud.

<u>Secure Communication Between Environments</u>

The on-premises data center and public cloud must communicate through a secure connection

The agency can use:

- VPN (Virtual Private Network): Creates an encrypted connection between the government data center and the cloud.
- Dedicated private connection: Provides a secure and reliable connection between the two environments.
- Firewalls: Allow only authorized network traffic.

Identity Management

Identity management ensures that only authorized users can access government systems and data.

The agency should use a central identity management system to manage users across both on-premises and cloud environments.

Important security methods include:

- Single Sign-On (SSO): Allows users to access multiple approved applications using one login.
- Multi-Factor Authentication (MFA): Requires an additional verification step, such as an OTP or authentication app.
- Role-Based Access Control (RBAC): Gives users access based on their job role.
- Least Privilege: Users should only get the permissions they actually need.
- Regular access reviews: Remove access when an employee leaves or changes their role.
- Identity federation: Allows users to securely access cloud services using their existing organization identity.

Encryption

Encryption is important for protecting sensitive government data.

The agency should use:

- Encryption at rest: Protects data stored in databases, servers, and cloud storage.
- Encryption in transit: Protects data while it moves between users, on-premises systems, and the cloud.
- Key management: Encryption keys should be stored and managed securely.

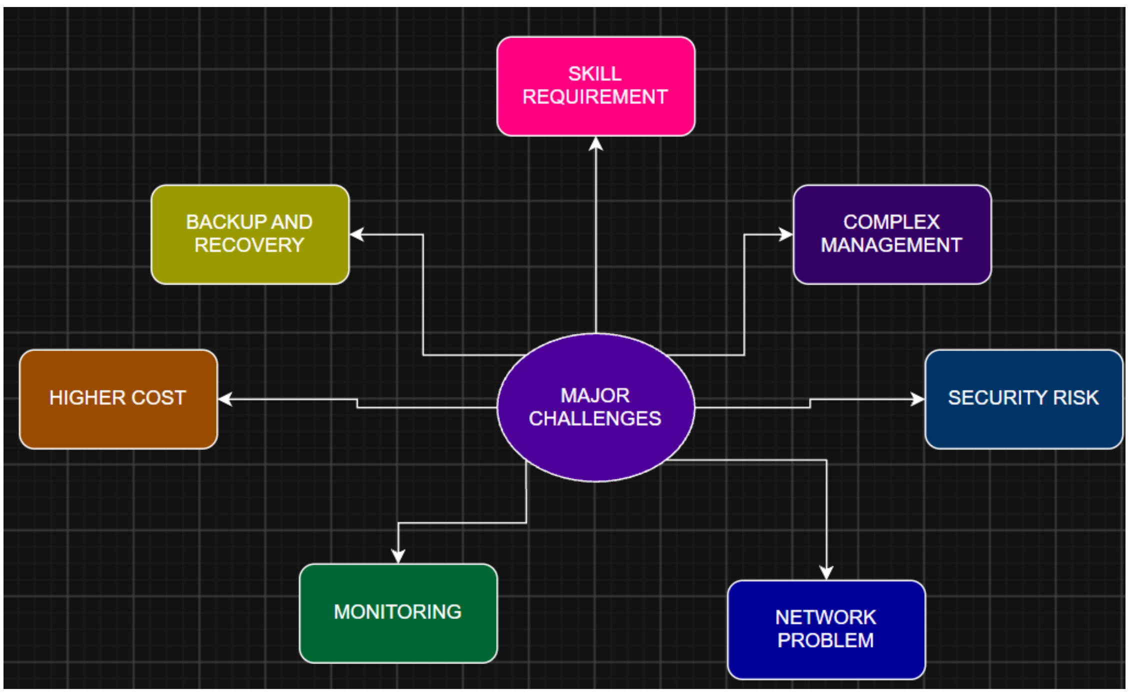
Even if someone gains unauthorized access to encrypted data, it will be difficult for them to understand the information without the encryption key.

Challenges in Managing Hybrid Cloud

Managing a hybrid cloud environment can be challenging because the agency has to manage both on-premises and cloud systems.

Some major challenges are:

- Complex management: Managing two different environments requires more effort.
- Security risks: Both environments must be protected from cyberattacks.
- Data synchronization: Keeping data consistent between cloud and on-premises systems can be difficult.
- Network problems: A slow or failed connection can affect communication between environments.
- Compliance: The agency must ensure that cloud services follow government regulations.
- Monitoring: It can be difficult to monitor all systems from one place.
- Higher cost: Maintaining on-premises infrastructure along with cloud services can increase expenses.
- Skill requirements: Staff need knowledge of both cloud and traditional IT systems.
- Integration issues: Applications running in different environments may not always work together easily.
- Backup and recovery: Data must be backed up properly across both environments.




- ❖ A hybrid cloud is a suitable solution for a government agency because it allows sensitive data and applications to remain on-premises, while non-sensitive applications can use the flexibility of the public cloud. A secure VPN or private connection can connect both environments.

5. A SaaS provider needs to ensure high availability and fault tolerance for its application. The system must continue operating even if one region goes down. Propose a multi-region deployment strategy in the cloud. Explain how data replication and failover mechanisms should be configured. Discuss monitoring and alerting strategies for proactive issue detection.

Answer:

A SaaS application may have thousands or millions of users, so it should be available all the time. If one cloud region goes down because of a server failure, network problem, or natural disaster, users should still be able to access the application. To achieve this, the SaaS provider can use a multi-region cloud deployment.

Multi-Region Deployment

The application can be deployed in two or more different cloud regions.

- Region 1 can be the main region.
- Region 2 can act as a backup or secondary region.
- Region 3 can also be used for additional availability and disaster recovery.

The application should run in multiple servers in each region. A global load balancer or traffic management service can direct users to the nearest healthy region.

Data Replication

Data should be copied between different regions so that the application can continue working even if one region becomes unavailable.

The provider can use:

- Automatic data replication between regions.
- Database replication to keep copies of important data.
- Object storage replication for files and documents.
- Regular backups as an additional safety measure.

Failover Mechanism

Failover means automatically moving users and services to another healthy region when the main region fails.

The system can work as follows:

- The monitoring system checks the health of each region.
- If Region 1 becomes unavailable, the system detects the failure.
- The global load balancer stops sending new users to Region 1.
- User traffic is automatically redirected to Region 2.
- The application continues running from Region 2.
- Once Region 1 is fixed, it can be brought back into service.

This process should be automated as much as possible to reduce downtime.

Monitoring and Alerting

The SaaS provider should continuously monitor the application and cloud infrastructure.

The following should be monitored:

- Server health.
- CPU and memory usage.
- Network performance.
- Application response time.
- Database health.

Alerts should be created when unusual problems are detected.

- For example, if the application response time suddenly increases or a region becomes unavailable, the monitoring system can immediately send an alert to the IT team through email, SMS, or messaging applications.
- The team can then take action before the problem becomes serious.

Health Checks

- The system should perform regular health checks on servers and applications.
- If a server or region fails the health check, the load balancer should automatically stop sending traffic to it and redirect users to a healthy region.
- This helps prevent users from accessing a failed service.
- A SaaS provider can achieve high availability by deploying its application across multiple cloud regions. A global load balancer can distribute user traffic, while data replication keeps copies of important data in different regions.
- If one region fails, automatic failover can redirect users to another healthy region. This multi-region approach provides high availability, fault tolerance, better performance, and reduced downtime for the SaaS application.

Benefits of Multi-Region Deployment

A multi-region setup provides: